

# 👉 XAI Lecture 10

## Sanity Checks for Saliency Maps & OpenXAI

FAMAF - UNC

First Semester 2026



**eXplainable**  
Artificial Intelligence

## Disclaimer ---

This course is based on

### **Explainable Artificial Intelligence**

From Simple Predictors to Complex Generative Models

Spring 2023, Harvard University

<https://interpretable-ml-class.github.io/>

6 Nov 2020

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## Sanity Checks for Saliency Maps

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<sup>†</sup>University of California Berkeley

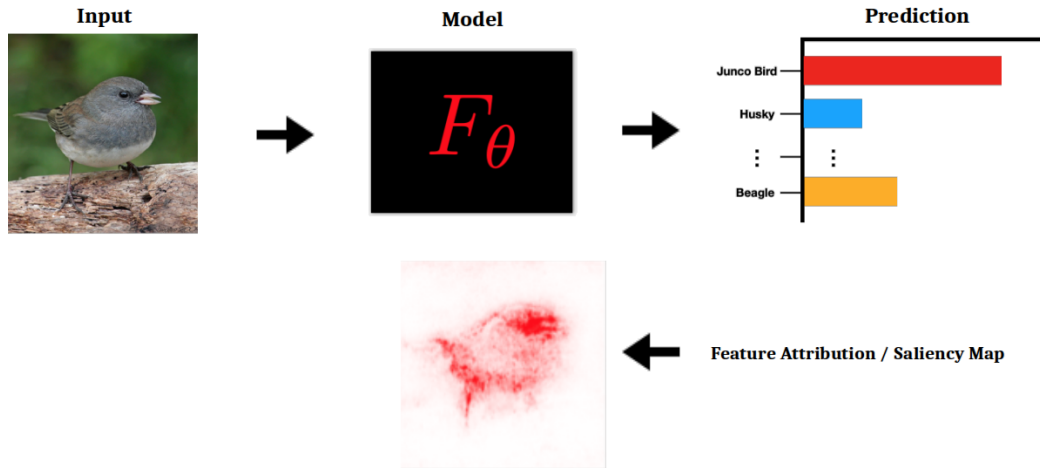
### Abstract

Saliency methods have emerged as a popular tool to highlight features in an input deemed relevant for the prediction of a learned model. Several saliency methods have been proposed, often guided by visual appeal on image data. In this work, we propose an actionable methodology to evaluate what kinds of explanations a given method can and cannot provide. We find that reliance solely on visual assessment

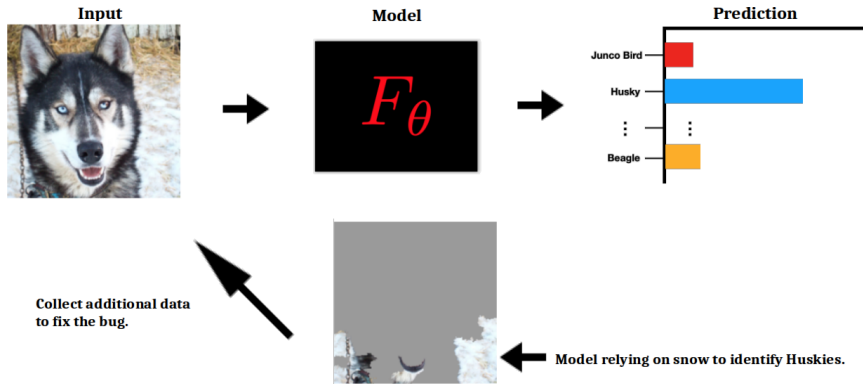
(Adebayo et al. 2018)

## 🍷 Feature Attributions / Saliency Maps ---

What parts of the input are 'most important' for the model prediction?



## 🍷 Identifying Shortcuts ---



A **shortcut** is a specific type of bias — the model learns a spurious correlation that works well on training data but fails to generalize. Instead of learning the relevant feature (the dog), it learns an easier, correlated signal (the snow).

## 🍷 Feature Attribution / Saliency Maps ---

**Feature attribution method:** assigns an output 'relevance' score to each dimension of the input.

$$F : \mathbb{R}^d \rightarrow \mathbb{R}^c \quad \text{Model}$$

---

$$F_i : \mathbb{R}^d \rightarrow \mathbb{R} \quad \text{class specific logit}$$

An attribution method produces a map  $E : \mathbb{R}^d \rightarrow \mathbb{R}^d$  of the same shape as the input, where each value  $E_j(x)$  indicates how much dimension  $j$  contributed to the prediction of class  $i$ .

## 🍷 Input-Gradient / Saliency / Gradient ---

### Input-Gradient:

$$\nabla_x F_i(x) \in \mathbb{R}^d$$

- Same dimension as the input
- Gradient w.r.t. **Input** for class **Logit**  $i$

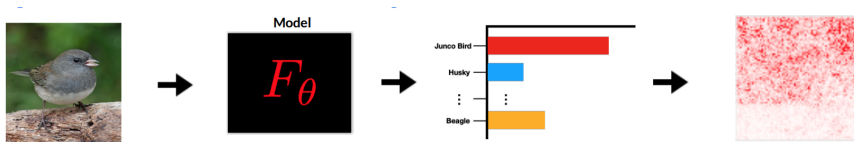
**Intuitively**,  $\nabla_x F_i(x)$  answers:

*"if I perturb input dimension  $j$  slightly, how much does the predicted score for class  $i$  change?"*

Large values indicate dimensions the model is most sensitive to near  $x$ .

(Baehrens et al. 2010; Simonyan, Vedaldi, and Zisserman 2014)

## 🍷 Input-Gradient / Saliency / Gradient - Demo ---



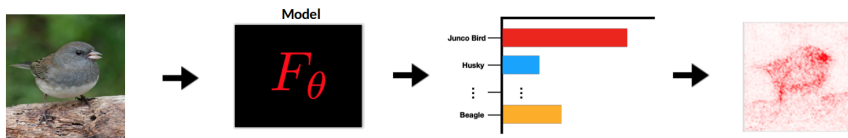
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## 🍷 Integrated Gradients ---

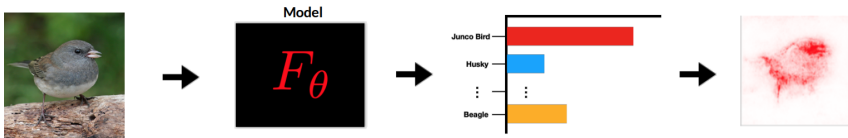


$$(x - \tilde{x}) \times \int_{\alpha=0}^1 \frac{\partial F(\tilde{x} + \alpha \times (x - \tilde{x}))}{\partial x}$$

**Path integral:** 'sum' of interpolated gradients from **Baseline input**  $\tilde{x}$  to  $x$ .

(Sundararajan, Taly, and Yan 2017)

## 🍷 SmoothGrad ---



$$\frac{1}{N} \sum_i^N \nabla_{(x+\epsilon)} F_i(x + \epsilon)$$

Average of gradients computed on input  $x$  perturbed with **Gaussian noise**  $\epsilon$ .

(Smilkov et al. 2017)

## 🍷 Guided Backprop: "Modified Backprop" ---



activation:

$$f_i^{l+1} = \text{relu}(f_i^l) = \max(f_i^l, 0)$$

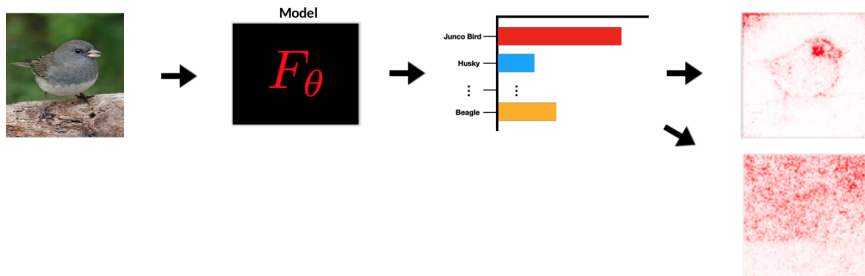
backpropagation:

$$R_i^l = (f_i^l > 0) \cdot R_i^{l+1}, \quad R_i^{l+1} = \frac{\partial f^{out}}{\partial f_i^{l+1}} R_i^l = (f_i^l > 0) \cdot \boxed{(R_i^{l+1} > 0)} \cdot R_i^{l+1}$$

guided backpropagation:

Additional gate: only backpropagate **positive** relevance signals

## 🍷 Are Saliency Maps Reliable? ---



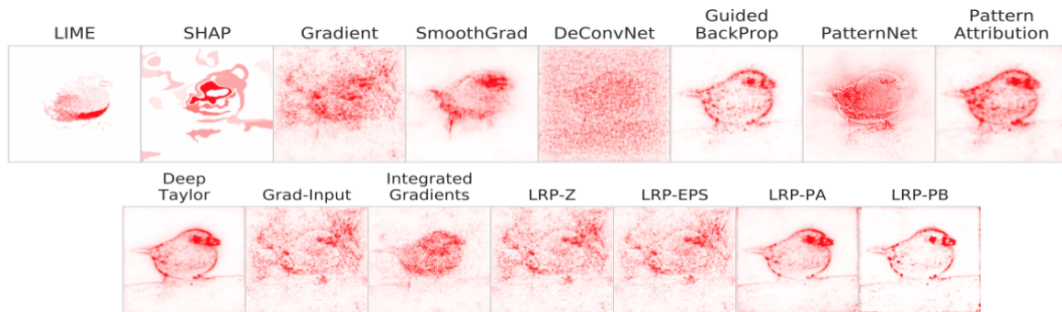
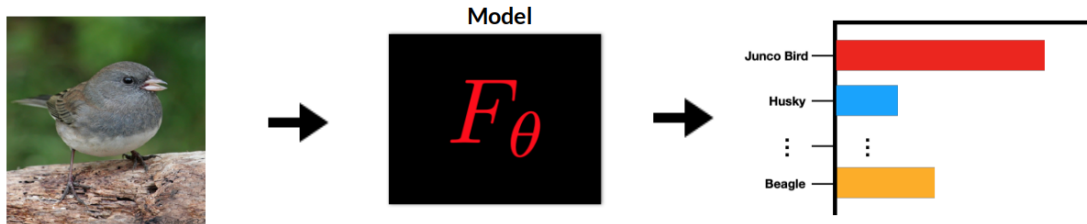
Both maps are produced by the **same method** on the **same input** — but one comes from a trained model and the other from a **randomly initialized** network.

- **Top:** explanation from a **trained** model.
- **Bottom:** explanation from a **randomly initialized** model.

Can you tell which is which?

*Visual inspection alone is not a reliable criterion for evaluating saliency methods.*

## 🍷 Recap: Many Saliency Methods ---



## Recap: Additional Methods ---

- **Class Activation Mapping** (Zhou et al. 2016)
- **Meaningful Perturbation** (Fong et al. 2017)
- **RISE** (Petsuik et al. 2018)
- **Extremal Perturbations** (Fong & Patrick 2019)
- **DeepLift** (Shrikumar et al. 2018)
- **Expected Gradients** (Erion et al. 2019)
- **GradCAM / Guided GradCAM** (Selvaraju et al. 2016)
- **Occlusion** (Zeiler et al. 2014)
- **Prediction Difference Analysis** (Gu et al. 2019)
- **Internal Influence** (Leino et al. 2018)

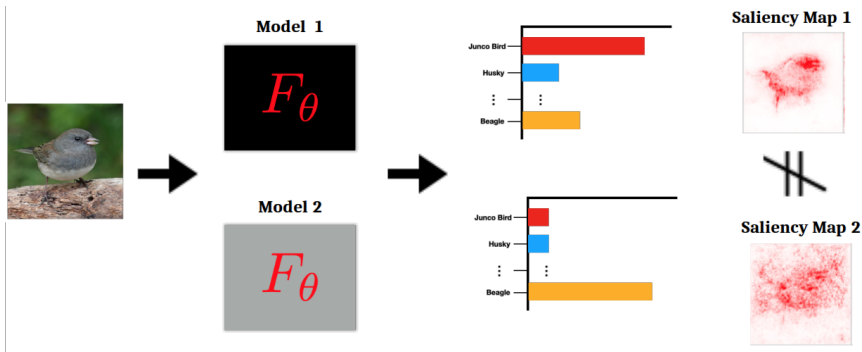
See for additional methods: Samek & Montavon et al. 2020

## 'Sanity Checks' ---

Intuitive 'principles' that an attribution method should satisfy.

- **'Model' Faithfulness:** is the 'explanation' sensitive to model parameters?
  - ▶ Test: change the model weights and measure corresponding change in explanation.
  - ▶ Operationalize by reinitialization of model weights.
- **Data Faithfulness:** is the attribution sensitive to training data?
  - ▶ Test: change training label and measure corresponding change in explanation.
  - ▶ Operationalize by randomization labelling in training data.

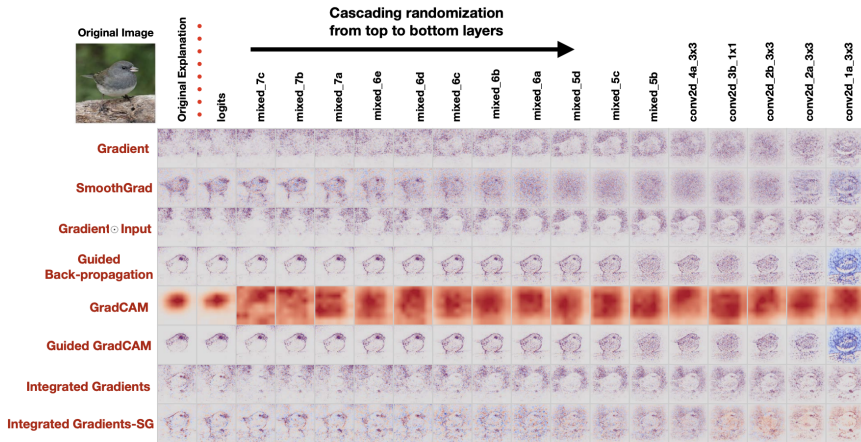
## 🍷 Sensitivity to Model Parameters ---



If the parameter settings of the model change, the saliency map should change.

- **Model 1**: trained normally
- **Model 2**: randomly initialized (or randomized top layers)
- Saliency maps should **differ** between models

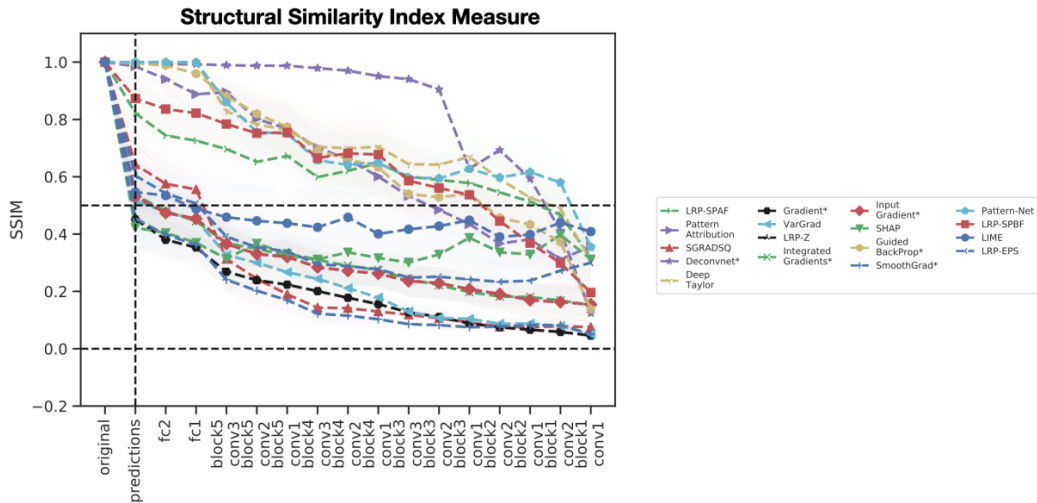
# 🍷 Sensitivity: Cascading Randomization ---



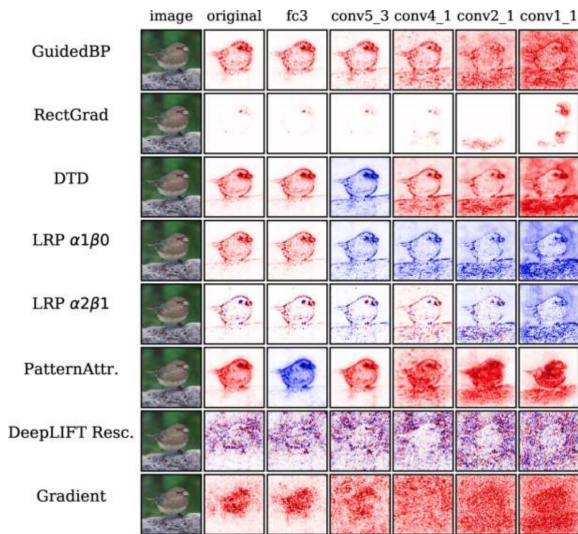
A method that looks the same regardless of the model's parameters cannot be explaining what the model learned.



# Sensitivity to Model Parameters: SSIM Results ---



## 🍷 Modified BackProp Approaches ---



Backprop-based methods that modify gradients at each layer collapse to a **rank-1 matrix** — their output is dominated by the **input structure**, not by the model's learned weights.

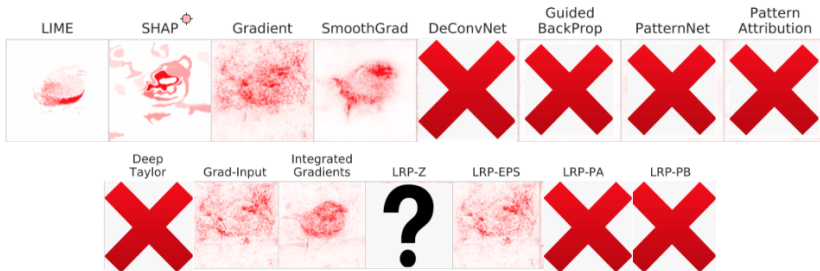
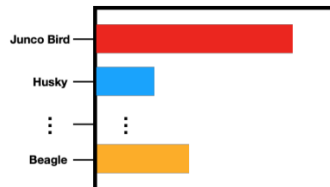
**This is why these methods still produce recognizable maps even when the model is completely random: they are recovering the input, not explaining the model.**

*(Sixt et al. 2020, Theorem 1)*

# 🍷 Recap: Which Method Should You Use? ---



Model



## Some Takeaways ---

- Identified certain classes of feature attribution methods that are **invariant to higher layer weights**
- 'Sanity Checks' are actually **weak** requirements, i.e., does **not** tell you whether a method is effective

## Some Objections 1/2 ---

Causal reframing suggests that sanity checks results might be **task specific**.

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### Revisiting Sanity Checks for Saliency Maps

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Gal Yona  
Weizmann Institute of Science

Daniel Greenfeld  
Jethor Energy Research

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## On the Relationship Between Explanation and Prediction: A Causal View

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## Some Objections 2/2 ---

Where you choose to perform randomization matters, and perhaps the **weight randomization is not the best approach**.

### Shortcomings of Top-Down Randomization-Based Sanity Checks for Evaluations of Deep Neural Network Explanations

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Grégoire Montavon<sup>4,5</sup>[0000-0001-7243-6186], Klaus-Robert Müller<sup>5,6,7,8</sup>, and Wojciech  
Samek<sup>3,5,6</sup>[0000-0002-6283-3265]

## 🍲 More Recent Observations: Spurious Features ---

Beyond faithfulness, it is unclear whether these feature attribution methods are effective for **model debugging**.

### **Do Feature Attribution Methods Correctly Attribute Features?**

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# More Recent Observations ---

## Do Input Gradients Highlight Discriminative Features?

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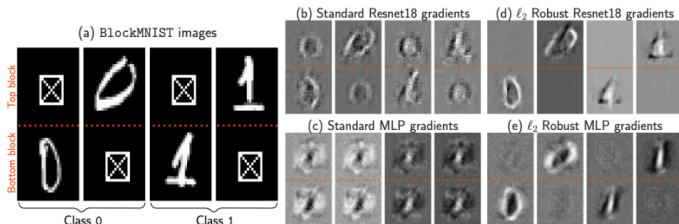
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## RETHINKING THE ROLE OF GRADIENT-BASED ATTRIBUTION METHODS FOR MODEL INTERPRETABILITY

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- **BlockMNIST**: each image has two blocks — the discriminative one is fixed by class position. A good saliency method should focus on the correct block.
- **Key finding**: saliency quality depends not only on the attribution method, but also on **how the model was trained** — robustly trained models produce cleaner, more discriminative gradients.

## 🍷 Parting Thoughts ---

Feature attribution is still important for applications, however, additional work is needed to characterize the properties of DeepNN model training that will result in 'gradients' that capture discriminative signals.

] 13 Mar 2024

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## OpenXAI: Towards a Transparent Evaluation of Post hoc Model Explanations

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Chirag Agarwal<sup>1</sup>, Dan Ley<sup>1</sup>, Satyapriya Krishna<sup>1</sup>, Eshika Saxena<sup>\*1</sup>, Martin Pawelczyk<sup>1</sup>, Nari  
Johnson<sup>4</sup>, Isha Puri<sup>\*1</sup>, Marinka Zitnik<sup>1</sup>, and Himabindu Lakkaraju<sup>1</sup>

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<sup>4</sup>Carnegie Mellon University

### Abstract

While several types of post hoc explanation methods have been proposed in recent literature, there is very little work on systematically benchmarking these methods. Here, we introduce OpenXAI, a comprehensive and extensible open-source framework for evaluating and benchmarking post hoc explanation methods.

(Agarwal et al. 2022)

# Overview ---

- Reliability pillars
- OpenXAI
- Is OpenXAI all you need?
- New directions

# 🍷 Applications of Saliency Methods 1/2 ---



## Natural images Fong et al. :

From: johnchad@triton.unm.edu (jchadwic)

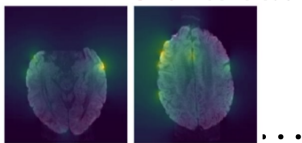
Subject: Another request for Darwin Fish

Organization: University of New Mexico, Albuquerque

Lines: 11

NNTP-Posting-Host: triton.unm.edu

## Text Ribeiro et al. 2016



## MRI brain scans

Agarwal et al. 2021

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## Audio

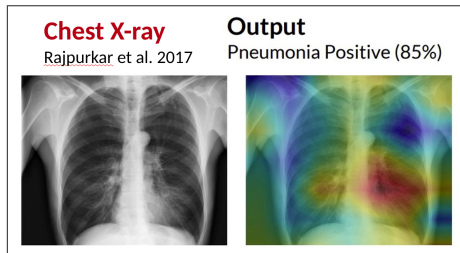
Becker et al. 2019



Saliency methods are used across a wide range of data modalities — the question of how to evaluate them is relevant in all of these settings.

- **Natural images** — highlighting relevant regions in a scene
- **MRI / medical imaging** — localizing diagnostically relevant areas
- **Text** — identifying influential words or tokens
- **Video** — attributing predictions to specific frames or regions
- **Audio** — localizing relevant time-frequency patterns in spectrograms

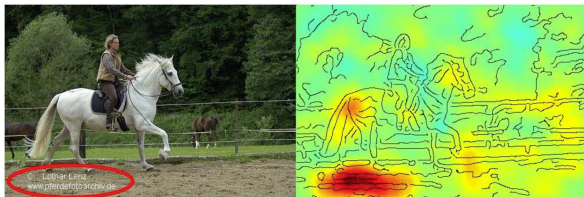
## 🍷 Applications of Saliency Methods 2/2 ---



In high-stakes domains, unreliable explanations can have serious consequences.

- **Medical diagnosis** — a model predicts Pneumonia (85%), but is it looking at the right region of the X-ray?
- **Detecting biases** — a horse classifier was found to rely on a **photographer's watermark** rather than the animal itself.

**Detecting biases** Lapuschkin et al. 2016



*A visually compelling explanation is not necessarily a correct one — this is precisely what makes evaluation frameworks like OpenXAI necessary.*

## 🍷 How Do We Evaluate the Reliability of Explanation Methods? ---

A reliable explanation method should satisfy three key properties (pillars):

- **Faithful** — the explanation accurately reflects the model's true behavior.
- **Stable** — similar inputs should produce similar explanations.
- **Fair** — explanation quality should not vary across demographic groups.

## Pillar 1: Faithfulness ---

**An explanation is faithful if it accurately reflects the model's true behavior.**

- **Intuitively:** if we hide the features the explanation marks as unimportant, the model's prediction should not change much.
- **Definition.** Given an input  $x$  and its explanation  $E_x$ :

$$\frac{1}{N} \sum_N \|f(x) - f(t(E_x, x))\|_2$$

where  $t(E_x, x)$  masks the features deemed unimportant by  $E_x$ .

A **lower value** indicates higher faithfulness.

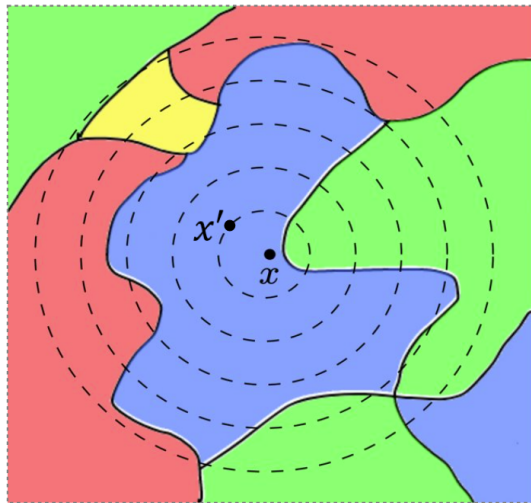
## 🍷 Pillar 2: Stability ---

**An explanation is stable if similar inputs produce similar explanations.**

- **Intuitively:** a small perturbation to the input  $x$  should not drastically change what the explanation highlights.
- **Definition.** Given an input  $x$  and a perturbed counterpart  $x'$ , the explanation  $E_x$  is stable if:

$$D(E_x, E_{x'}) \leq \delta$$

where  $D$  measures the distance between the two explanations and  $\delta$  is a small tolerance threshold.



## Pillar 3: Counterfactual Fairness ---

**An explanation is fair if changing a protected attribute only affects the explanation to the extent that it affects the model's prediction.**

- **Intuitively:** if the model treats two individuals the same regardless of their protected attribute, their explanations should also be the same.
- **Definition.** Given a feature vector  $x$  and its protected attribute perturbation  $x^S$ , an explanation  $E_x$  preserves counterfactual fairness if:

$$D(E_x, E_{x^S}) \propto f(x) - f(x^S)$$

The difference between explanations should be **proportional** to the difference in predictions. If the model's output didn't change, the explanation shouldn't change either.

How do we **pick** an explanation method  
from the XAI landscape?

# How do we pick an explanation method from the XAI landscape?

**Short answer:** it depends. **Longer answer:** it really depends.

**The XAI landscape offers dozens of methods** — but there is no universal winner. The best method depends on your model, your data, and what “reliable” means in your specific context.

## This is exactly what OpenXAI is for:

- OpenXAI provides an **automated end-to-end pipeline** that simplifies and standardizes the evaluation of post hoc explanation methods
- OpenXAI promotes **transparency and reproducibility** in benchmarking explanation methods

C. Agarwal et al., OpenXAI: Towards a Transparent Evaluation of Model Explanations, NeurIPS Datasets and Benchmark Track'2022

<https://github.com/AI4LIFE-GROUP/OpenXAI>

## OpenXAI's Key Components ---

- A flexible **synthetic data generator** and a collection of 7 real-world datasets, **16 pre-trained models**, and **6 state-of-the-art explanation methods**
- Open-source implementations of **22 quantitative metrics** for evaluating faithfulness, stability (robustness), and fairness of explanation methods
- **First-ever public XAI leaderboards** to benchmark explanation methods

## XAI-Ready Dataloaders and Models ---

OpenXAI includes 7 real-world datasets (finance, healthcare, criminal justice) and a synthetic data generator — all preprocessed and ready to use out of the box.

```
from openxai import Dataloader
loader_train, loader_test = Dataloader.return_loaders(
    data_name='german',
    download=True,
)
inputs, labels = iter(loader_test).next()
```

Pre-trained models (ANNs and logistic regression) are also available, so you can start benchmarking explanation methods without training anything yourself.

```
from openxai import LoadModel
model = LoadModel(data_name='german', ml_model='ann')
```

## OpenXAI Explainers ---

OpenXAI provides ready-to-use implementations of six state-of-the-art feature attribution methods: LIME, SHAP, Vanilla Gradients, Gradient  $\times$  Input, SmoothGrad, and Integrated Gradients.

```
from openxai import Explainer
exp_method = Explainer(method='LIME')
explanations = exp_method.get_explanations(model, X=inputs, y=labels)
```

This makes it easy to swap methods and compare their outputs under identical conditions.

## OpenXAI Explainers - API ---

Any custom method can be integrated by extending the `Explainer` class and implementing a single method:

```
@abstractmethod
def get_explanations(self, model, X: torch.Tensor, y: torch.Tensor):
    """
    Generate explanations for given input/s.
    Parameters: model, X (m x n tensor), y (labels)
    Returns: torch.Tensor (explanation vector/matrix)
    """
    pass
```

## 🍷 OpenXAI's Evaluation ---

OpenXAI provides implementations and ready-to-use APIs for a set of **22 quantitative metrics** proposed by prior research to evaluate the faithfulness, stability, and fairness of explanation methods.

```
from openxai import Evaluator
metric_evaluator = Evaluator(inputs, labels, model, explanations)
score = metric_evaluator.eval(metric='RIS')
```

# OpenXAI's Leaderboard 1/2 ---

OpenXAI provides a public leaderboard to compare explanation methods across datasets, models, and evaluation metrics — making it easy to identify which method works best for a given setting.

## Explore Leaderboards

German Credit

### Faithfulness

| Method                              | <u>FA</u>  | <u>RA</u> | <u>SA</u> | <u>SRA</u> | <u>RC</u> | <u>PRA</u> | <u>PGI</u> | <u>PGU</u> |
|-------------------------------------|---------------------------------------------------------------------------------------------|-----------|-----------|------------|-----------|------------|------------|------------|
| <a href="#">Vanilla Gradient</a>    | 0.950                                                                                       | 0.950     | 0.846     | 0.846      | 1.000     | 1.000      | 0.149      | 0.174      |
| <a href="#">SmoothGrad</a>          | 0.950                                                                                       | 0.950     | 0.606     | 0.606      | 1.000     | 1.000      | 0.202      | 0.124      |
| <a href="#">Integrated Gradient</a> | 0.950                                                                                       | 0.950     | 0.846     | 0.846      | 1.000     | 1.000      | 0.148      | 0.173      |
| <a href="#">LIME</a>                | 0.938                                                                                       | 0.810     | 0.938     | 0.814      | 0.998     | 0.989      | 0.156      | 0.169      |
| <a href="#">Gradient x Input</a>    | 0.785                                                                                       | 0.161     | 0.382     | 0.071      | 0.890     | 0.875      | 0.171      | 0.161      |
| <a href="#">SHAP</a>                | 0.130                                                                                       | 0.007     | 0.112     | 0.006      | -0.053    | 0.488      | 0.135      | 0.180      |

# 🍷 OpenXAI's Leaderboard 2/2 ---

## Explore Leaderboards

German Credit

### Faithfulness

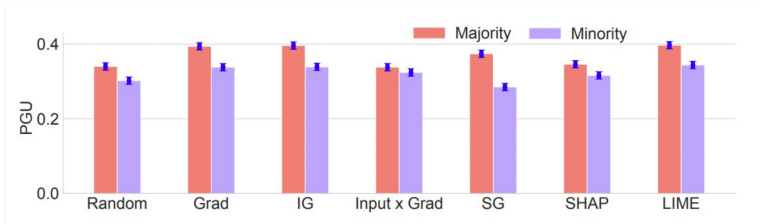
| Method              | FA $\uparrow$ | RA    | SA    | SRA   | RC     | PRA   | PGI   | PGU   |
|---------------------|---------------|-------|-------|-------|--------|-------|-------|-------|
| Vanilla Gradient    | 0.950         | 0.950 | 0.846 | 0.846 | 1.000  | 1.000 | 0.149 | 0.174 |
| SmoothGrad          | 0.950         | 0.950 | 0.606 | 0.606 | 1.000  | 1.000 | 0.202 | 0.124 |
| Integrated Gradient | 0.950         | 0.950 | 0.846 | 0.846 | 1.000  | 1.000 | 0.148 | 0.173 |
| LIME                | 0.938         | 0.810 | 0.938 | 0.814 | 0.998  | 0.989 | 0.156 | 0.169 |
| Gradient x Input    | 0.785         | 0.161 | 0.382 | 0.071 | 0.890  | 0.875 | 0.171 | 0.161 |
| SHAP                | 0.130         | 0.007 | 0.112 | 0.006 | -0.053 | 0.488 | 0.135 | 0.180 |

The table shows **faithfulness metrics** for six methods on the **German Credit** dataset:

- **FA, RA, SA, SRA, PRA, RC** — agreement between explanation and ground truth.
- **PGI / PGU** — prediction gap when masking important / unimportant features.

**Key observation:** gradient-based methods consistently outperform SHAP — but this may not hold across all settings.

## 🍷 Exploring the Landscape Using OpenXAI ---



No single method wins on all dimensions:

- **LIME** produces more faithful explanations (+24.9%).
- **SmoothGrad** achieves 63.2% higher stability (RRS) — but shows the **largest fairness gap** between majority and minority groups.

The chart shows PGU scores by demographic group. A gap between majority (red) and minority (purple) indicates unequal explanation quality across groups.

**A method that is faithful and stable may still be unfair**

## 🍷 Is OpenXAI All You Need? ---

- How to benchmark different **non-perturbation-based** explanation methods?
- Benchmarking explanations on **other modalities**:
  - ▶ Vision (Quantus): <https://quantus.readthedocs.io/en/latest/>
  - ▶ NLP (e-ViL): <https://github.com/maximek3/e-ViL>
  - ▶ Graphs (GraphXAI): <https://github.com/mims-harvard/GraphXAI>

C. Agarwal et al., Evaluating Explainability for Graph Neural Networks, Nature Scientific Data'2023

M. Kayser et al., e-ViL: A Dataset and Benchmark for Natural Language Explanations in Vision-Language Tasks, ICCV'2021

## New Directions ---

- Training models using **Explanation Feedbacks**
- **Differentiable Explainable Curricula** for RL Agents
- Learning **Hierarchical and Multi-modal Explanations**

## 🍷 Should You Use OpenXAI Today? ---

| Aspect               | Assessment                                        |
|----------------------|---------------------------------------------------|
| Conceptual framework | Still valid and well-cited                        |
| Metrics coverage     | Comprehensive (faithfulness, stability, fairness) |
| Active maintenance   | Limited since 2022–2023                           |
| Compatibility        | May lag behind current SHAP/LIME versions         |

### Alternatives worth knowing:

- **Quantus** — actively maintained, strong on faithfulness & robustness metrics
- **Captum** (PyTorch) — deep integration with neural networks
- **CARLA** — focused on counterfactual explanations

**Bottom line:** OpenXAI is a solid reference for *understanding evaluation methodology*, but for hands-on projects, consider more actively maintained tools.

# Thank You!

## References --- |

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